

Notes: Leary & Roberts (JF, 2014)

Do Peer Firms Affect Corporate Financial Policy?

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1 Big picture: what the paper is doing

1.1 Research question

The paper asks whether corporate financial policies (especially leverage and financing) exhibit *peer effects*:

Do firms adjust their capital structures because other firms in the same peer group adjust theirs?

The core empirical challenge is that firms in the same industry can look similar for many reasons that have nothing to do with imitation: shared technology, shared demand shocks, common regulation, common lenders, etc. The paper’s goal is to separate these confounding “correlated effects” from a genuine “endogenous” peer effect.

1.2 Main contribution

The paper implements a clean peer-effects identification strategy in corporate finance by instrumenting peers’ financial-policy choices using *peers’ idiosyncratic stock-return shocks* (constructed from an asset-pricing factor model), and then quantifies the *economic multiplier* implied by the estimated peer effect.

1.3 Empirical punchlines (as reflected in the tables)

- Reduced-form: peers’ exogenous equity shocks predict a firm’s leverage and issuance behavior (Table IV) in a way similar to the firm’s own equity shocks.
- Structural (2SLS): the estimated endogenous peer effect is positive and economically meaningful for leverage (Table VII): firms partially move with peers’ leverage choices.
- Economic implication: the peer effect generates *amplification* (social-multiplier style) and *spillovers* across firms; these are stronger when peer groups are smaller (Table VIII).
- Directionality: the evidence is consistent with “followers” responding to “leaders” but not vice versa (Table X).

2 Data objects and notation used throughout

2.1 Sample and peer groups

- Annual Compustat panel, nonfinancial and nonutility firms, 1965–2008.
- Baseline peer groups are defined by three-digit SIC industry codes.
- “Peer-firm averages” exclude the i th firm from the industry-year mean (leave-one-out construction).

2.2 Outcome variables (financial policy)

Let y_{ijt} denote a financial-policy outcome for firm i in industry j in year t . The paper studies, among others:

- market leverage and book leverage (levels),
- changes in leverage,
- equity issuance and debt issuance indicators (linear probability models).

2.3 Controls: firm-specific and peer-average characteristics

Let X_{ijt-1} be the firm-specific characteristics (lagged one year), such as: firm size (log sales), market-to-book, profitability (EBITDA/assets), and tangibility (net PPE/assets). Let $\bar{X}_{-ijt-1}$ be the peer-group (industry-year) average of these characteristics excluding firm i .

Table I reports summary statistics for both the *firm-specific* variables and their *peer averages*, in both levels and first differences.

3 The baseline peer-effects model

3.1 Empirical model

The paper posits a linear peer-effects model for financial policy:

$$y_{ijt} = \alpha + \beta \bar{y}_{-ijt} + \gamma' \bar{X}_{-ijt-1} + \lambda' X_{ijt-1} + \delta' \mu_j + \phi' \nu_t + \varepsilon_{ijt}, \quad (1)$$

where:

- $\bar{y}_{-ijt}$ is the contemporaneous peer-firm average outcome (leave-one-out),
- μ_j and ν_t are industry and year fixed effects,
- β captures the *endogenous peer effect* (peer actions),
- γ captures *contextual peer effects* (peer characteristics).

3.2 Why the contemporaneous peer outcome matters

Using $\bar{y}_{-ijt}$ contemporaneously (rather than lagged) makes it harder for a story based purely on sequential learning to explain the results mechanically, and it pushes the specification toward capturing a near-simultaneous “move-with-the-group” component of policy choice.

3.3 The econometric problem: reflection / simultaneity

Equation (1) is not directly identified by OLS because $\bar{y}_{-ijt}$ is endogenous: if firm i responds to peers, then peers respond to firm i as well. This is the corporate-finance version of the Manski (1993) “reflection problem.”

4 Identification problem and reduced form: derivations

This section reproduces (and explains) the key algebra that motivates the need for an instrument.

4.1 Population version (suppressing year fixed effects)

For notational convenience, consider a population version of the model that conditions on an industry effect μ_j and treats the peer outcome through the conditional mean $\mathbb{E}(y \mid \mu_j)$:

$$y = \alpha + \beta \mathbb{E}(y \mid \mu_j) + \gamma' \mathbb{E}(X \mid \mu_j) + \lambda' X + \delta' \mu_j + \varepsilon. \quad (2)$$

4.2 Mean regression given (X, μ_j)

Taking conditional expectations of (2) yields:

$$\mathbb{E}(y \mid X, \mu_j) = \alpha + \beta \mathbb{E}(y \mid \mu_j) + \gamma' \mathbb{E}(X \mid \mu_j) + \lambda' X + \delta' \mu_j. \quad (3)$$

4.3 Equilibrium condition for $\mathbb{E}(y \mid \mu_j)$

Now take expectations of (3) with respect to X conditional on μ_j :

$$\mathbb{E}(y \mid \mu_j) = \alpha + \beta \mathbb{E}(y \mid \mu_j) + \lambda' \mathbb{E}(X \mid \mu_j) + \gamma' \mathbb{E}(X \mid \mu_j) + \delta' \mu_j. \quad (4)$$

4.4 Solving for $\mathbb{E}(y \mid \mu_j)$

Rearrange (4):

$$(1 - \beta) \mathbb{E}(y \mid \mu_j) = \alpha + (\gamma + \lambda)' \mathbb{E}(X \mid \mu_j) + \delta' \mu_j.$$

Assuming $\beta \neq 1$ (and in practice thinking of $|\beta| < 1$ for stability), the unique solution is:

$$\mathbb{E}(y \mid \mu_j) = \frac{\alpha}{1 - \beta} + \left(\frac{\gamma + \lambda}{1 - \beta} \right)' \mathbb{E}(X \mid \mu_j) + \left(\frac{\delta}{1 - \beta} \right)' \mu_j. \quad (5)$$

4.5 Reduced form

Plugging (5) into (3) yields the reduced-form model:

$$\mathbb{E}(y \mid X, \mu_j) = \alpha^* + \gamma^{*'} \mathbb{E}(X \mid \mu_j) + \delta^{*'} \mu_j + \lambda^{*'} X, \quad (6)$$

with composite (reduced-form) parameters:

$$\alpha^* = \frac{\alpha}{1 - \beta}, \quad \gamma^{*'} = \left(\frac{\beta\lambda + \gamma}{1 - \beta} \right)', \quad \delta^{*'} = \left(\frac{\delta}{1 - \beta} \right)', \quad \lambda^{*'} = \lambda'.$$

4.6 Key implication

Estimating (6) does *not* identify β because β is folded into the composite parameters. To identify the endogenous peer effect β in the structural equation (1), we need an *exogenous shifter of peers' outcomes* $\bar{y}_{-ijt}$.

5 The instrument: peer firms' idiosyncratic equity shocks

Table I
Summary Statistics

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents means, standard deviations (*SD*), and medians for variables in levels and first differences. Peer Firm Averages denotes variables constructed as the average of all firms within an industry-year combination, excluding the i^{th} observation. Industries are defined by three-digit SIC code. Firm-Specific Factors denotes variables corresponding to firm i 's value in year t .

	Levels			First Differences		
	Mean	Median	<i>SD</i>	Mean	Median	<i>SD</i>
<i>Peer Firm Averages</i>						
Book Leverage (Total Debt/Book Assets)	0.238	0.229	0.094	0.004	0.003	0.031
Market Leverage	0.274	0.262	0.137	0.006	0.004	0.058
Log(Sales)	5.085	4.932	1.278	0.091	0.094	0.119
Market-to-Book	1.362	1.201	0.650	-0.032	-0.019	0.310
EBITDA/Assets	0.108	0.120	0.070	-0.002	-0.001	0.031
Net PPE/Assets	0.317	0.270	0.172	-0.002	-0.002	0.020
<i>Firm-Specific Factors</i>						
Book Leverage (Total Debt/Book Assets)	0.238	0.217	0.196	0.004	-0.000	0.098
Market Leverage (Total Debt/Market Assets)	0.274	0.216	0.246	0.006	0.000	0.123
Log(Sales)	5.085	5.018	2.172	0.091	0.089	0.357
Market-to-Book	1.362	0.966	1.244	-0.032	-0.006	0.829
EBITDA/Assets	0.108	0.129	0.155	-0.002	0.000	0.104
Net PPE/Assets	0.317	0.271	0.217	-0.002	-0.002	0.060
<i>Industry Characteristics</i>						
No. of Firms per Industry-Year	13.217	8.000	18.344			
Total No. of Industries	217					
<i>Sample Characteristics</i>						
Observations	80,279					
Firms	9,126					

Table II
Stock Return Factor Regression Results

The sample consists of monthly returns for all nonfinancial, nonutility firms in the intersection of the annual Compustat and monthly CRSP databases between 1965 and 2008. The table presents mean factor loadings and adjusted R^2 s from the regression

$$R_{ijt} = \alpha_{ijt} + \beta_{ijt}^M(RM_t - RF_t) + \beta_{ijt}^{IND}(\bar{R}_{-ijt} - RF_t) + \eta_{ijt},$$

where R_{ijt} is the return to firm i in industry j during month t , $(RM_t - RF_t)$ is the excess return on the market, and $(\bar{R}_{-ijt} - RF_t)$ is the excess return on an equal-weighted industry portfolio excluding firm i 's return, where industries are defined by three-digit SIC code. The regression is estimated for each firm on a rolling annual basis using historical monthly returns data from the CRSP database. We require at least 24 months of historical data and use up to 60 months of data in the estimation. Expected returns are computed using the estimated factor loadings and realized factor returns 1 year hence. Idiosyncratic returns are computed as the difference between realized and expected returns.

	Mean	Median	<i>SD</i>
α_{it}	0.008	0.007	0.017
β_{it}^M	0.399	0.422	0.803
β_{it}^{IND}	0.616	0.535	0.567
Obs. per Regression	59	60	5
Adjusted R^2	0.228	0.207	0.170
Avg. Monthly Return	0.013	0.000	0.182
Expected Monthly Return	0.015	0.014	0.090
Idiosyncratic Monthly Return	-0.002	-0.011	0.174

5.1 Construction of idiosyncratic returns (factor model)

The instrument is built from firm-level *idiosyncratic* stock returns, after removing market-wide and industry-wide components. For each firm i (in industry j), the paper estimates a rolling factor regression:

$$R_{ijt} = \alpha_{it} + \beta_{it}^M(RM_t - RF_t) + \beta_{it}^{IND}(\bar{R}_{-i,j,t} - RF_t) + \eta_{ijt}, \quad (7)$$

where:

- R_{ijt} is the firm monthly return,
- $(RM_t - RF_t)$ is the market excess return,
- $(\bar{R}_{-i,j,t} - RF_t)$ is the excess return on an equal-weighted industry portfolio excluding firm i ,

- η_{ijt} is the idiosyncratic component.

Estimation uses at least 24 months and up to 60 months of historical monthly returns in a rolling window.

5.2 Expected vs. idiosyncratic returns

Given estimated factor loadings, define the expected return:

$$\hat{r}_{ijt} = \hat{\alpha}_{it} + \hat{\beta}_{it}^M(RM_t - RF_t) + \hat{\beta}_{it}^{IND}(\bar{R}_{-i,j,t} - RF_t), \quad (8)$$

and the idiosyncratic return:

$$\hat{\eta}_{ijt} = R_{ijt} - \hat{r}_{ijt}. \quad (9)$$

5.3 Peer-firm average equity shock (the instrument)

Aggregate idiosyncratic returns to the annual level (details are in the paper's implementation) and define the *leave-one-out* peer average:

$$\bar{\eta}_{-ijt} = \frac{1}{N_{jt} - 1} \sum_{\substack{k \in j \\ k \neq i}} \hat{\eta}_{kjt}, \quad (10)$$

where N_{jt} is the number of firms in industry j in year t .

In the structural (2SLS) estimations, the instrument is the one-period-lagged peer shock $\bar{\eta}_{-ij,t-1}$.

5.4 What Figure 1 is showing

Figure 1 plots the empirical distribution of $\bar{\eta}_{-ijt}$ under different peer-group definitions (3-digit, 2-digit, 1-digit SIC):

- The distribution is centered near zero (consistent with idiosyncratic shocks having mean zero).
- As peer groups become broader (1-digit), the distribution becomes more concentrated around zero because averaging over more firms reduces variance.

5.5 Why this instrument can be credible

- **Exogeneity:** idiosyncratic peer shocks are, by construction, orthogonal to market and industry return components, and should be unrelated to firm i 's fundamentals after controlling for i 's own shock and fixed effects.
- **Relevance:** peer equity shocks should affect peers' leverage mechanically (via market value) and through financing choices (equity issuance/repurchase, debt issuance), thereby shifting $\bar{y}_{-ijt}$.

5.6 Evidence on exogeneity (Table III)

Table III regresses the peer-firm average equity shock on firm characteristics (and their leads). The estimates are economically small and mostly statistically insignificant, and the one-period-ahead characteristics do not predict the shock. This supports viewing $\bar{\eta}_{-ijt}$ as a plausible exogenous shifter after controls.

6 Reduced-form results: what moves with peer shocks? (Table IV)

The paper first estimates reduced-form relations of the type:

$$y_{ijt} = a + b\bar{\eta}_{-ij,t-1} + c'\bar{X}_{-ij,t-1} + d'X_{ij,t-1} + \mu_j + \nu_t + u_{ijt},$$

interpreting b as a *reduced-form* peer-shock effect.

Table III
Peer Firm Return Shock Properties

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents estimated coefficients and t -statistics robust to heteroskedasticity and within-firm dependence in parentheses. The dependent variable is the average peer firm equity return shock. All independent variables are in levels and are either contemporaneous with or a one-period-lead relative to the dependent variable, as indicated at the top of the columns. Firm-Specific Factors denotes variables corresponding to firm i 's value in year t . Peer Firm Average Characteristics are peer firm averages of the same variables listed under firm-specific factors in the table: log of sales, the market-to-book ratio, the ratio of EBITDA to assets, and the ratio of net PPE to assets. Peer firm averages are constructed as the average of all firms within an industry-year combination, excluding the i^{th} observation. Industries are defined by three-digit SIC code. Statistical significance at the 5% and 1% levels is denoted by * and **, respectively.

	Peer Firm Average Equity Shock	
	Contemporaneous Independent Vars.	1-Period-Lead Independent Vars.
<i>Firm-Specific Factors</i>		
Log(Sales)	-0.000 (-0.565)	-0.000 (-0.334)
Market-to-Book	-0.001 (-1.444)	0.000 (0.104)
EBITDA/Assets	-0.009* (-2.336)	-0.000 (-0.048)
Net PPE/Assets	0.008 (1.934)	0.004 (0.994)
Peer Firm Average Characteristics	Yes	Yes
Firm i Equity Return Shock	Yes	Yes
Industry Fixed Effects	Yes	Yes
Year Fixed Effects	Yes	Yes
Obs.	80,279	80,119
Adj. R^2	0.128	0.127

Table IV
Peer Effects in Financial Policy: Reduced-Form Estimates

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). Both panels present OLS estimated coefficients and t -statistics robust to heteroskedasticity and within-firm dependence in parentheses. The dependent variable is indicated at the top of columns. All independent variables are lagged 1 year and are in levels or first differences (Δ) for consistency with the dependent variable indicated at the top of the columns. The exception is stock returns, which are in level form across all specifications. Equity (Debt) Issuance is equal to one if Net Stock (Debt) Issuances normalized by lagged book assets is greater than 1%. The last column of Panel A isolates the subsample of observations in which either an equity or debt issuance, but not both, occurred. In Panel B, the change in market leverage is the dependent variable in all specifications, which include all firm-specific and peer firm averages used in Panel A as control variables. Stock Return Controls includes firm i 's lagged and contemporaneous total stock return. Additional Control Variables includes lagged firm-specific and peer firm averages for changes in cash flow volatility, a dividend payer indicator, Altman's (1968) Z -score, Graham's (2000) marginal tax rate, capital investment, R&D expenditures, and SG&A expenditures as well as the intraindustry standard deviation of leverage. See Appendix A for complete variable definitions. Polynomials of Control includes quadratic and cubic terms of all independent variables other than industry average leverage. Contemporaneous Controls replaces the lagged firm-specific and peer firm averages control variables with contemporaneous values. Statistical significance at the 5% and 1% levels is denoted by * and **, respectively.

	Market Leverage (1)	Book Leverage (2)	Δ Market Leverage (3)	Δ Book Leverage (4)	Equity Issuance (5)	Debt Issuance (6)	(Issuers) Debt Issuance (7)
Panel A: Financial Policy							
<i>Peer Firm Averages</i>							
Equity Shock	-0.024** (-4.965)	-0.016** (-3.997)	-0.020** (-6.312)	-0.008** (-3.109)	0.021* (2.440)	-0.029* (-2.469)	-0.034* (-2.313)
Log(Sales)	-0.002 (-0.869)	-0.002 (-0.994)	0.030** (5.941)	0.009* (2.398)	-0.008* (-2.295)	0.008 (1.790)	0.014** (2.810)
Market-to-Book	0.001 (0.314)	0.001 (0.426)	-0.001 (-0.431)	-0.001 (-0.343)	0.024** (4.817)	0.030** (5.079)	-0.011 (-1.505)
EBITDA/Assets	0.045 (1.574)	0.118** (4.666)	-0.026 (-1.475)	-0.021 (-1.377)	0.038 (0.909)	0.357** (7.130)	0.306** (4.840)
Net PPE/Assets	0.078** (2.700)	0.025 (1.005)	0.057* (1.972)	0.050* (2.392)	0.066 (1.836)	-0.008 (-0.192)	-0.006 (-0.131)
<i>Firm-Specific Factors</i>							
Equity Shock	-0.008** (-5.812)	-0.002 (-1.624)	0.001 (1.317)	-0.002 (-1.844)	0.062** (21.199)	0.020** (6.282)	-0.052** (-12.219)
Log(Sales)	0.010** (9.260)	0.010** (10.812)	0.016** (8.071)	0.006** (3.274)	-0.012** (-9.353)	0.014** (11.251)	0.024** (13.173)
Market-to-Book	-0.053** (-43.272)	-0.014** (-11.384)	-0.002** (-3.500)	-0.002** (-3.057)	0.077** (35.226)	0.005* (2.421)	-0.071** (-28.260)
EBITDA/Assets	-0.308** (-28.937)	-0.231** (-20.865)	-0.033** (-5.400)	-0.024** (-3.511)	-0.258** (-17.546)	-0.041** (-2.689)	0.161** (7.991)
Net PPE/Assets	0.161** (12.007)	0.195** (16.169)	0.068** (6.851)	0.053** (5.685)	0.042** (3.094)	0.185** (12.220)	0.086** (4.422)
Industry Fixed Effects	Yes	Yes	No	No	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	80,279	80,279	80,279	80,279	80,279	80,279	35,363
Adj. R^2	0.31	0.20	0.10	0.01	0.16	0.05	0.27

6.1 Table IV, Panel A: broad financial-policy outcomes

Table IV Panel A shows that lagged peer equity shocks predict:

Table IV—Continued

	Δ Market Leverage					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel B: Change in Leverage Robustness Tests						
<i>Peer Firm Averages</i>						
Equity Shock	−0.014** (−4.866)	−0.019** (−5.522)	−0.014** (−3.102)	−0.019** (−6.131)	−0.015** (−5.318)	−0.021** (−6.487)
Peer Firm Averages	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Specific Factors	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	No	No	No	No	No	No
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Stock Return Controls	Yes	No	No	No	No	No
Additional Control Variables	No	Yes	No	No	No	No
Bank $\times$ Market Return Effects	No	No	Yes	No	No	No
Lagged Dependent Variable	No	No	No	Yes	No	No
Contemporaneous Controls	No	No	No	No	Yes	No
Polynomials of Controls	No	No	No	No	No	Yes
Obs.	80,279	69,578	33,674	80,230	80,119	80,279
Adj. R^2	0.26	0.10	0.11	0.10	0.18	0.10

- lower market and book leverage (negative coefficients),
- lower changes in leverage (negative coefficients),
- higher probability of equity issuance and lower probability of debt issuance (sign pattern consistent with equity shocks relaxing financing constraints / increasing equity attractiveness).

A useful way to read Panel A is: *peer shocks are associated with firms moving in the same “direction” as they do when they themselves experience equity shocks.*

6.2 Table IV, Panel B: robustness for Δ market leverage

Panel B shows that the negative reduced-form relationship between peer shocks and Δ market leverage is stable when:

- adding or changing controls (including contemporaneous controls),
- adding lagged dependent variables,
- allowing nonlinearities (polynomials),
- altering return-control specifications.

This reinforces that the reduced-form signal is not driven by a narrow modeling choice.

7 Ruling out correlated-effects stories

7.1 Customer–supplier tests (Table V)

The concern: peer shocks might proxy for omitted industry fundamentals.

Table V shifts the shock source away from the focal firm’s industry by using shocks to *customers of peer firms* in different industries. Because the customer industry differs from firm i ’s industry, this design reduces the scope for a same-industry unobserved factor to drive the results. Panel A

Table V
Customer-Supplier Tests

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). Panel A presents OLS estimates using peer groups defined as the subset of firms in the same industry as firm i that satisfy the following two criteria: (1) they have customers in an industry different from firm i , and (2) their customers are not customers of firm i . Panel B presents OLS estimates in which we replace each customer from the analysis in Panel A with a randomly selected noncustomer in the same industry as the customer. The average shock to the randomly selected noncustomers is then used in place of the shock to the customer. We perform the random selection and OLS estimation 100 times to obtain a distribution of estimated coefficients and t -statistics on the noncustomer equity return shocks. All specifications include firm-specific and peer firm averages for firm size, profitability, tangibility, and the market-to-book ratio. Statistical significance at the 5% and 1% levels is denoted by * and **, respectively.

	Market Leverage (1)	Δ Market Leverage (2)	Issue Debt (3)
Panel A: Customer Return Shocks			
Avg. Peer Customer Equity Shock	-0.012* (-2.398)	-0.011** (-3.535)	-0.036* (-2.027)
Industry Avg. Equity Return	-0.044** (-10.898)	0.009** (3.553)	-0.042** (-3.211)
<i>Peer Firm Averages</i>			
Log(Sales)	0.004* (2.339)	0.018** (4.978)	0.002 (0.589)
EBITDA/Assets	0.062** (5.031)	0.002 (0.278)	0.130** (3.431)
Market-to-Book	0.001 (0.759)	-0.002* (-2.294)	-0.001 (-0.216)
Net PPE/Assets	-0.013 (-0.842)	0.055** (2.942)	-0.049 (-1.256)
<i>Firm-Specific Factors</i>			
Log(Sales)	0.009** (8.271)	0.024** (12.219)	0.027** (11.015)
EBITDA/Assets	-0.176** (-20.583)	-0.035** (-6.440)	0.134** (6.143)
Market-to-Book	-0.039** (-39.523)	-0.003** (-5.795)	-0.062** (-24.398)
Net PPE/Assets	0.199** (13.153)	0.063** (5.366)	0.148** (5.701)
Equity Shock	-0.009** (-6.719)	0.004** (3.878)	-0.044** (-9.187)
Industry Fixed Effects	Yes	No	Yes
Year Fixed Effects	Yes	No	Yes
Obs.	54,599	52,222	21,410
Adj. R^2	0.28	0.08	0.26

	Percentiles					
	Mean	5	25	50	75	95
Panel B: Placebo Tests						
<i>Coefficient Estimates</i>						
Market Leverage	0.004	-0.002	0.001	0.004	0.006	0.010
Δ Market Leverage	0.001	-0.004	-0.001	0.000	0.003	0.006
Issue Debt	0.001	-0.019	-0.006	0.001	0.007	0.021
<i>Peer Effect t-stats</i>						
Market Leverage	1.496	-0.831	0.283	1.537	2.499	4.033
Δ Market Leverage	0.414	-2.738	-0.887	0.274	1.704	3.475
Issue Debt	0.080	-2.016	-0.638	0.111	0.763	2.398

still finds that these customer shocks correlate with leverage and debt issuance in the predicted direction.

Panel B performs placebo tests by randomizing the “customer” assignment within the customer industry. The placebo distribution of coefficients is centered near zero, supporting that Panel A is not a mechanical artifact.

7.2 Do shocks matter only through peers’ actions? (Table VI)

Table VI provides a simple but powerful diagnostic: it sorts observations by (i) the peer equity shock and (ii) the peer leverage change, and then looks at the firm’s leverage change.

The striking pattern is that the firm’s leverage change is strongly monotonic in *peer leverage changes* (columns), and much less sensitive to the peer equity shock once peer leverage changes are held fixed (rows). This supports the view that the equity shock is mainly valuable as an *instrument that shifts peers’ policies*, not as a direct proxy for peer characteristics.

8 Structural peer effects: two-stage least squares (Table VII)

8.1 2SLS implementation

To estimate β in (1), the paper treats the peer average outcome $\bar{y}_{-ijt}$ as endogenous. A stylized representation of the two stages is:

First stage (relevance).

$$\bar{y}_{-ijt} = \pi_0 + \pi_1 \bar{\eta}_{-ij,t-1} + \pi_2' \bar{X}_{-ij,t-1} + \pi_3' X_{ij,t-1} + \mu_j + \nu_t + v_{ijt}. \quad (11)$$

Second stage (structural peer effect).

$$y_{ijt} = \alpha + \beta \hat{\bar{y}}_{-ijt} + \gamma' \bar{X}_{-ij,t-1} + \lambda' X_{ij,t-1} + \mu_j + \nu_t + \varepsilon_{ijt}. \quad (12)$$

8.2 The coefficients in Table VII

Table VII reports 2SLS coefficients *scaled by the standard deviation of the corresponding regressor*. So, for example, the coefficient on the peer average dependent variable in the market leverage regression can be read as:

A one-standard-deviation increase in peers' market leverage is associated (causally, under the IV assumptions) with about a 0.10 standard-deviation increase in firm i 's market leverage.

8.3 Main structural findings from Table VII

- **Leverage peer effects:** β is positive and significant for leverage levels and changes (columns (1)–(4)).
- **Issuance outcomes:** evidence is stronger for equity issuance and for debt issuance within the issuer subsample; the broad debt-issuance indicator is weaker.
- **First-stage strength:** the instrument strongly predicts peer outcomes (bottom row in Table VII); reported test statistics indicate that weak-instrument concerns are limited in the main leverage specifications.

9 Economic magnitude: amplification and spillovers (Appendix B and Table VIII)

The peer-effects model implies that changing one firm's characteristics can affect: (i) its own policy more than "direct" effects would suggest (amplification), and (ii) other firms' policies (spillovers).

Table VI
Leverage Changes by Peer Firm Equity Shock and Peer Firm Leverage Change

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents average market leverage changes for 25 groups of observations. The groups are formed by the intersection of quintiles based on: (1) peer firm average equity return shocks lagged 1 year and (2) peer firm average change in market leverage. The column labeled "5 - 1" presents the difference in means between columns 5 and 1. The row labeled "5 - 1" presents the difference in means between rows 5 and 1. t -statistics robust to heteroskedasticity and within-firm dependence are in parentheses. Statistical significance at the 1% level is denoted by **.

Lagged Peer Firm	Peer Firm Avg Leverage Change Quantiles					
	1 (Low)	2	3	4	5 (High)	5 - 1
1 (Low)	-0.033** (-14.176)	-0.008** (-4.026)	0.007** (3.158)	0.021** (9.857)	0.062** (29.059)	0.095**
2	-0.044** (-18.302)	-0.014** (-6.574)	0.007** (4.348)	0.020** (9.665)	0.062** (26.558)	0.106**
3	-0.042** (-18.606)	-0.014** (-7.284)	-0.000 (-0.253)	0.023** (13.002)	0.066** (25.001)	0.108**
4	-0.047** (-22.532)	-0.013** (-7.628)	0.003 (1.566)	0.017** (8.202)	0.066** (28.275)	0.114**
5 (High)	-0.046** (-27.376)	-0.024** (-11.640)	0.006** (2.644)	0.016** (7.327)	0.062** (24.489)	0.108**
5 - 1	-0.014**	-0.016**	-0.000	-0.005	-0.000	

Table VII
Peer Effects in Financial Policy: Structural Estimates

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents two-stage least squares (2SLS) estimated coefficients scaled by the corresponding variable's standard deviation, and t -statistics robust to heteroskedasticity and within-firm dependence in parentheses. The dependent variable is indicated at the top of each column. The endogenous variable is the peer firm average of the dependent variable. The instrument is the one-period-lagged peer firm average equity return shock. Peer Firm Averages denotes variables constructed as the average of all firms within an industry-year combination, excluding the i^{th} observation. Industries are defined by three-digit SIC code. Firm-Specific Factors denotes variables corresponding to firm i 's value in year t . All variables are in levels or first differences as indicated at the top of the columns. All independent variables, including the instrument but excluding the endogenous variable, are lagged 1 year relative to the dependent variable unless otherwise specified. Equity (Debt) Issuance are indicator variables equal to one if Net Stock (Debt) Issuances normalized by lagged book assets is greater than 1%. The last column of Panel A isolates the subsample of observations in which either an equity or debt issuance, but not both, occurred. Statistical significance at the 5% and 1% levels is denoted by * and **, respectively.

	Market Leverage (1)	Book Leverage (2)	Δ Market Leverage (3)	Δ Book Leverage (4)	Equity Issuance (5)	Debt Issuance (6)	(Issuers) Debt Issuance (7)
<i>Peer Firm Averages</i>							
Dependent Variable	0.100** (4.803)	0.100** (3.573)	0.077** (5.431)	0.045* (2.566)	0.046* (2.423)	-0.631 (-1.249)	0.105* (2.230)
Log(Sales)	-0.011** (-2.719)	-0.015** (-3.111)	-0.004* (-2.440)	-0.001 (-0.777)	-0.003 (-0.599)	0.095 (1.387)	0.001 (0.120)
Market-to-Book	0.032** (4.403)	0.011** (2.791)	0.001 (1.463)	0.002 (1.602)	-0.002 (-0.278)	0.092 (1.555)	0.017 (1.410)
EBITDA/Assets	0.021** (4.618)	0.020** (4.949)	0.002* (2.416)	0.001 (1.151)	0.008* (2.244)	0.102 (1.632)	0.006 (0.744)
Net PPE/Assets	-0.013 (-1.659)	-0.033** (-2.858)	-0.001 (-1.788)	-0.001 (-1.251)	0.007 (1.095)	0.083 (1.200)	-0.001 (-0.144)
<i>Firm-Specific Factors</i>							
Log(Sales)	0.021** (9.113)	0.021** (10.435)	0.005** (6.541)	0.002** (2.640)	-0.026** (-9.161)	0.035** (7.432)	0.051** (12.410)
Market-to-Book	-0.067** (-42.570)	-0.018** (-11.257)	-0.002** (-3.661)	-0.002** (-3.081)	0.095** (34.880)	0.015 (1.871)	-0.097** (-27.751)
EBITDA/Assets	-0.048** (-28.700)	-0.037** (-20.769)	-0.003** (-5.128)	-0.002** (-3.280)	-0.040** (-17.645)	0.005 (0.514)	0.026** (7.661)
Net PPE/Assets	0.034** (11.281)	0.042** (15.105)	0.004** (5.955)	0.003** (4.680)	0.009** (2.940)	0.040** (8.654)	0.018** (4.137)
Equity Shock	-0.003** (-4.468)	-0.000 (-0.484)	0.002** (3.030)	-0.000 (-0.713)	0.032** (21.071)	0.007 (1.535)	-0.028** (-12.039)
<i>First-Stage Instrument</i>							
Peer Firm Avg Equity Shock	-0.033** (-13.126)	-0.015** (-7.350)	-0.015** (-8.713)	-0.005** (-4.383)	0.074** (19.430)	0.009 (1.530)	-0.091** (-8.146)
<i>Industry Fixed Effects</i>							
Industry Fixed Effects	Yes	Yes	No	No	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	80,279	80,279	80,279	80,279	80,279	80,279	34,578

9.1 Matrix representation (Appendix B)

Fix an industry-year with N firms, and stack outcomes in a vector $y = (y_1, \dots, y_N)'$. Let Q be the $N \times N$ matrix with zeros on the diagonal and ones elsewhere (so Qy gives the sum of others' outcomes). Appendix B rewrites the model in matrix form:

$$y = \frac{\beta}{N-1} Qy + X\lambda + \frac{1}{N-1} QX\gamma + Z\delta + \varepsilon, \quad (13)$$

where X contains exogenous variables that appear both as firm-specific and peer averages, and Z contains other exogenous controls (fixed effects, etc.).

Solving (13):

$$y = \left(I - \frac{\beta}{N-1} Q \right)^{-1} \left(X\lambda + \frac{1}{N-1} QX\gamma + Z\delta + \varepsilon \right). \quad (14)$$

9.2 Closed-form marginal effects (Appendix B)

Consider an exogenous regressor x_{lm} (the m th characteristic of firm l). Appendix B derives the total derivative of y_i with respect to x_{lm} as:

$$\frac{\partial y_i}{\partial x_{lm}} = \begin{cases} \lambda_m \left(1 + \frac{\beta^2}{(N-1+\beta)(1-\beta)} \right) + \gamma_m \left(\frac{\beta}{(N-1+\beta)(1-\beta)} \right), & i = l, \\ \lambda_m \left(\frac{\beta}{(N-1+\beta)(1-\beta)} \right) + \gamma_m \left(\frac{1}{(N-1+\beta)(1-\beta)} \right), & i \neq l. \end{cases} \quad (15)$$

9.3 Interpreting the components: amplification and spillovers

Equation (15) separates three intuitive pieces:

- **Amplification term (own effect multiplier):**

$$1 + \frac{\beta^2}{(N-1+\beta)(1-\beta)}.$$

This exceeds 1 when $\beta \neq 0$ and is larger when peer groups are smaller (small N), because each firm has more influence on the peer average.

- **Spillover 1 (actions spillover via λ):**

$$\frac{\beta}{(N-1+\beta)(1-\beta)}.$$

This loads the firm-specific channel λ_m into other firms' outcomes.

- **Spillover 2 (characteristics spillover via γ):**

$$\frac{1}{(N-1+\beta)(1-\beta)}.$$

This transmits peer-characteristics effects across firms.

9.4 What Table VIII adds

Table VIII plugs the estimated β (from Table VII) into (15) and reports:

- the *scaled* direct coefficients ($\lambda \times \sigma_x$ and $\gamma \times \sigma_x$),
- the implied derivatives for three peer-group sizes (small, medium, large),
- the implied amplification and spillover magnitudes.

The main message is that amplification and spillovers are economically nontrivial in small peer groups and attenuate mechanically as N grows.

Table VIII
Exogenous Variable Derivatives, Marginal Effects, and Leverage Multipliers

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents 2SLS estimated coefficients scaled by the corresponding variable's standard deviation, and t -statistics robust to heteroskedasticity and within-firm dependence in parentheses from a regression of market leverage on peer firm leverage, peer firm characteristics, and firm-specific factors. All variables are in levels. Peer Firm Averages denotes variables constructed as the average of all firms within an industry-year combination, excluding the i th observation. Industries are defined by three-digit SIC code. Firm-Specific Factors denotes variables corresponding to firm i 's value in year t . Derivatives are computed for three peer groups differing in their size: small (10th percentile = three firms), medium (50th percentile = eight firms), and large (90th percentile = 36 firms). The derivative $\partial y_i / \partial x_{it}$ shows the change to the outcome of observation i (y_i) following a one unit change to variable x_{it} for observation i (x_{it}). The derivative $\partial y_i / \partial x_{it}$ shows the change to the outcome of observation i (y_i) following a one unit change to variable x_{it} for observation i (x_{it}). Both derivatives are scaled by the standard deviation of the corresponding x variable, (σ_x). Amplification Term is the multiplicative factor due to the peer effect action variable and is equal to $1 + (\frac{\partial y_i}{\partial x_{it}} \frac{\partial x_{it}}{\partial y_i})^{-1}$. The terms Spillover 1 and Spillover 2 are the additive factors due to the peer firm actions and characteristics, and are equal to $(\frac{\partial y_i}{\partial x_{it}} \frac{\partial x_{it}}{\partial y_i})^{-1}$ and $(\frac{\partial y_i}{\partial x_{it}} \frac{\partial x_{it}}{\partial y_i})^{-1}$, respectively. Statistical significance at the 5% and 1% levels is denoted by * and **, respectively.

Variable	Firm-Specific Factor		Peer Firm Average		Peer Group Size: Small		Peer Group Size: Medium		Peer Group Size: Large	
	Scaled Coeff ($\lambda \times \sigma_y$)	Scaled Coeff ($\lambda \times \sigma_y$)	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$	$\frac{\partial y_i}{\partial x_{it}} \times \sigma_x$
Log(Sales)	0.021**	-0.011**	0.018*	-0.004	0.020**	-0.002	0.021**	-0.000		
	(9.113)	(-2.719)	(2.964)	(-0.902)	(6.903)	(-0.500)	(8.202)	(-0.499)		
Market-to-Book	-0.067**	0.032**	-0.055**	0.016	-0.063**	0.006	-0.065**	0.002		
	(-42.570)	(4.403)	(-4.597)	(1.225)	(-13.751)	(1.178)	(-32.156)	(1.161)		
IBITDA/Assets	-0.048**	0.021**	-0.038**	0.015	-0.040**	0.005	-0.047**	0.002		
	(-28.700)	(4.618)	(-3.678)	(1.337)	(-11.231)	(1.285)	(-23.794)	(1.266)		
Net PPE/Assets	0.034**	-0.013	0.042**	0.011	0.036**	0.004	0.035**	0.001		
	(11.281)	(-1.609)	(5.298)	(1.257)	(8.968)	(1.241)	(10.984)	(1.235)		
Peer Firm Avg. Leverage (β)	0.727**									
	(4.803)									
Amplification Term			1.710		1.251		1.078			
			(1.092)		(1.051)		(1.037)			
Spillover 1			0.977		0.345		0.108			
			(1.414)		(1.346)		(1.322)			
Spillover 2			1.343*		0.474		0.148			
			(2.004)		(1.870)		(1.824)			

Table IX
Which Firms Mimic?

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents 2SLS estimated coefficients for the peer firm average market leverage change interacted with indicator variables identifying the lower and upper third of the within industry-year distribution of lagged values for firm-specific measures of whether the firm has a credit rating, whether the firm paid a dividend in year $t-1$, market share, profitability, market-to-book ratio, and the Whited-Wu (2006) Index of financial constraints. We exclude the middle third of the distribution for each of these regressions. The coefficient estimates are scaled by the corresponding variable standard deviation. The dependent variable is the change in market leverage ratio. All models are estimated by linear 2SLS where the endogenous variables are the peer firm average leverage ratio changes interacted with indicator variables, and the instruments are the one-period-lagged peer firm average idiosyncratic component of stock returns interacted with the same indicator variables. The table also presents the heteroskedasticity-corrected Cragg-Donald (1990) statistic testing for weak instruments (First-Stage Multivariate F -stat). Industries are defined by three-digit SIC code. All variables are in first differences except the instrument. All independent variables, including the instrument but excluding the endogenous variable, are lagged 1 year relative to the dependent variable unless otherwise specified. All test statistics are computed using standard errors that are robust to within-firm correlation and heteroskedasticity. Statistical significance at the 5% and 1% levels is denoted by * and **, respectively. F -stat statistical significance implying less than 15% or 10% size distortion is denoted by * and **, respectively.

	Credit Rating (2=Yes)	Dividend Payer (2=Yes)	Market Share (2=Large)	Profitability (2=High)	Market-to-Book (2=High)	Whited-Wu Index (2=Constrd)
Peer Firm Avg. Leverage Change × Group 1	0.073** (5.332)	0.061** (5.312)	0.050** (3.811)	0.056** (4.808)	0.054** (4.056)	0.031* (2.391)
Peer Firm Avg. Leverage Change × Group 2	0.027** (3.387)	0.046** (4.933)	0.041** (2.861)	0.055** (4.047)	0.056** (4.513)	0.053** (3.639)
First-Stage Multivariate F -stat	37.987**	38.149**	19.528**	23.618**	28.189**	21.061**
Peer Firm Average Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Specific Factors	Yes	Yes	Yes	Yes	Yes	Yes
Industry Fixed Effects	No	No	No	No	No	No
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	80,279	80,279	47,338	55,416	55,418	40,655

10 Heterogeneity: who mimics and who is mimicked? (Tables IX and X)

10.1 Which firms mimic? (Table IX)

Table IX estimates versions of the leverage-change specification where the peer leverage change is interacted with group indicators defined by firm characteristics: credit rating, dividend payer status, market share, profitability, market-to-book, and the Whited–Wu index of financial constraints. The peer variables and the instruments are interacted with the same indicators, and the middle third of continuous characteristics is excluded.

A reading of Table IX is:

- The peer effect is present across most splits (both groups generally show positive and significant coefficients).
- Some patterns suggest stronger mimicry where informational or institutional frictions are likely more severe (e.g., financially constrained firms per Whited–Wu show a larger peer coefficient).

10.2 Which firms are mimicked? (Table X)

Table X imposes a “leader–follower” structure:

- Firms are classified within an industry-year as “leaders” (upper third) and “followers” (middle and lower thirds) based on profitability, market share, or stock returns.
- Panel A regresses followers’ leverage changes on leaders’ average leverage changes (instrumenting with leaders’ lagged average idiosyncratic return shocks).
- Panel B flips the regression to test whether leaders respond to followers.

The results indicate that followers respond to leaders (positive and significant coefficients in Panel A), whereas leaders do not respond to followers (near-zero and insignificant coefficients in Panel B). This directional asymmetry is consistent with learning/reputation interpretations of peer effects.

Table X
Which Firms Are Mimicked?

The sample consists of all nonfinancial, nonutility firms in the annual Compustat database between 1965 and 2008 with nonmissing data for all analysis variables (see Appendix A). The table presents estimated coefficients scaled by the corresponding variable's standard deviation, and t -statistics robust to heteroskedasticity and within-firm dependence in parentheses. All models are estimated by linear 2SLS where the endogenous variable is the industry average leverage change and the instrument is the one-period-lagged industry average idiosyncratic component of stock returns. Industries are defined by three-digit SIC code. All specifications include one-period-lagged peer firm averages and firm-specific effects for the following characteristics: firm size, profitability, tangibility, and the market-to-book ratio. Firms are classified as either "Leaders" or "Followers" based on their within-industry-year ranking by: profitability, market share (sales as a fraction of industry sales), stock returns, and earnings growth. The table restricts attention to the subsample of firms in the middle and lower thirds of the within-industry-year distribution (i.e., Followers) of each classification variable and regresses their change in market leverage ratio on the average change in market leverage of firms in the upper third (i.e., Leaders), as well as the control variables indicated toward the bottom of the table. F -stat statistical significance implying less than 10% or 15% size distortion is denoted by * and **, respectively.

	Change in Market Leverage		
	Profitability	Market Share	Stock Return
Panel A: Do Follower Firms Respond to Leaders?			
Leader Firm Avg Leverage Change	0.076** (5.055)	0.084** (6.045)	0.087** (3.830)
First-Stage Univariate F -stat	62.913**	92.725**	34.530**
Peer Firm Average Characteristics	Yes	Yes	Yes
Firm Specific Factors	Yes	Yes	Yes
Industry Effects	No	No	No
Year Fixed Effects	Yes	Yes	Yes
Obs.	51,588	43,677	51,590
Panel B: Do Leader Firms Respond to Followers?			
Follower Firm Avg Leverage Change	-0.019 (-0.890)	-0.002 (-0.042)	-0.088 (-1.711)
First-Stage Univariate F -stat	17.688**	5.235*	6.021*
Peer Firm Average Characteristics	Yes	Yes	Yes
Firm Specific Factors	Yes	Yes	Yes
Industry Effects	No	No	No
Year Fixed Effects	Yes	Yes	Yes
Obs.	53,554	45,529	53,552

11 Conclusion

11.1 Summary

This paper concludes that **peer firms causally affect corporate financial policy**. In particular, firms adjust leverage and (especially) equity issuance in response to the contemporaneous financial policy choices of other firms in the same peer group (industry).

The main empirical problem is the "reflection" issue: firms and their peers co-move because they share common fundamentals, so a naive regression of a firm outcome on the peer average outcome does not identify a causal peer effect. The paper resolves this by introducing **plausibly exogenous peer shocks**—peer-firm idiosyncratic equity-return innovations—that shift peer firms' leverage/issuance choices, and then using these shocks as instruments in a 2SLS framework.

The central empirical conclusions are:

- (i) **Strong leverage peer effects.** The structural 2SLS estimates imply that a one-standard-deviation increase in peer-firm average leverage increases firm i 's leverage by about **10 percentage points** in both market and book leverage specifications (Table VII, columns (1)–(2): scaled coefficients ≈ 0.100). Peer effects are also present in *changes* in leverage (Table VII, columns (3)–(4): scaled coefficients ≈ 0.077 and 0.045).
- (ii) **Peer effects operate strongly through equity issuance.** Peer effects are statistically and economically important for equity issuance decisions (Table VII, column (5): scaled coefficient

≈ 0.046). In contrast, debt issuance peer effects are weak in the full sample but become economically meaningful when conditioning on financing activity (Table VII, columns (6)–(7)).

- (iii) **Peer effects generate amplification and spillovers (externalities).** Because the outcome depends on the peer average outcome, shocks to one firm propagate to others, and the total effect of any exogenous determinant is **larger than its partial derivative**. The implied amplification is economically large and depends on peer-group size: the paper reports that determinant shocks can be magnified by about **71%** in small peer groups (e.g., $N = 3$) but only about **8%** in large peer groups (e.g., $N = 26$).
- (iv) **Mimicking is concentrated among followers and constrained firms.** Peer sensitivity is stronger among firms that look like followers (smaller market share, lower profitability/growth), and among firms that appear more financially constrained (Table IX discussion). A direct leader–follower test shows that followers respond to leaders (Table X, Panel A: scaled coefficients ≈ 0.076 – 0.087), while leaders do not respond to followers (Table X, Panel B: coefficients near zero and statistically weak).

11.2 Why peers matter for leverage and issuance

The paper’s intuition is that **financial policy is chosen under uncertainty and frictions**, so firms rationally use information contained in peers’ observed actions.

Step 1: why leverage co-moves across firms. Leverage and issuance decisions reflect hard-to-measure objects: investment opportunities, risk, and financing conditions. If firms face model uncertainty or limited information about these objects, observing peers’ actions provides a signal about what is currently “appropriate” in the industry.

Step 2: why the causal channel is through *peer actions* rather than *peer characteristics*. Peer characteristics (average profitability, tangibility, etc.) can matter, but the evidence suggests that the dominant channel is **peer behavior itself**: peer shocks influence peers’ leverage/issuance, and firm i then responds to those peer choices. This is consistent with learning/reputation mechanisms where actions summarize private assessments and constraints.

Step 3: why the instrument is conceptually sensible. Peer-firm idiosyncratic equity-return shocks are designed to be:

- **relevant:** they shift peers’ financing choices (e.g., equity gains lower leverage mechanically and may facilitate equity issuance),
- **exogenous to firm i :** they are orthogonal to firm i ’s own unobservables after controlling for market and industry factors.

Thus, they move $\bar{y}_{-i,j,t}$ for reasons not mechanically tied to firm i 's latent determinants, enabling identification of β .

Step 4: why amplification is natural. If firm i responds to peers and peers respond to firm i , then any exogenous shift to one firm induces a cascade of responses within the peer group. This creates an equilibrium multiplier: the effect of a primitive determinant is amplified relative to a world with no peer feedback, and the strength of this multiplier is mechanically larger when the peer group is smaller.

11.3 Implications (what this changes in corporate finance)

The paper implies that capital structure is not a purely firm-level optimization outcome. Instead, leverage and issuance decisions exhibit **strategic interdependence** and **information externalities**. As a result:

- industry-wide leverage waves can reflect endogenous propagation rather than only common shocks,
- policy or financing shocks to a subset of firms can transmit to others through peer channels,
- empirically, researchers should treat peer-group outcomes as potentially endogenous equilibrium objects rather than controls.

11.4 Takeaway

Using peer firms' idiosyncratic equity-return shocks as instruments, Leary and Roberts (2014) show that peer actions causally affect leverage and equity issuance, producing economically large within-industry propagation and amplification effects that are strongest among follower and financially constrained firms.